# Neuro-Symbolic Biomass Prediction with Logical Tensor Networks

*Himal Garg | June 2026 | Technical Report*

## 1. Introduction & Objective

Accurate estimation of pasture biomass is essential for livestock management and feed budgeting in precision agriculture. Traditional methods require destructive sampling, which is time-consuming and impractical at scale. This project explores an automated, non-destructive approach that predicts five biomass components from pasture images and associated metadata.

The five target variables are: Dry Clover (g), Dry Dead (g), Dry Green (g), Dry Total (g), and Green Dry Matter - GDM (g). These targets are not independent: they obey known agronomic identities (e.g., Dry Total = Clover + Dead + Green). A standard neural network can learn to predict each target but may produce physically inconsistent outputs. A purely symbolic model can enforce the equations but cannot learn from raw image data.

Logical Tensor Networks (LTNs) bridge this gap. By grounding a neural regression model as a differentiable function within a first-order fuzzy logic framework, LTNs allow us to inject domain knowledge as soft logical axioms directly into the training loss. The model thus learns to be both perceptually accurate and logically consistent.

## 2. Experimental Setup

### 2.1 Dataset

The dataset consists of 357 unique pasture plot images, each associated with tabular metadata (NDVI, average plant height, geographical state, and species). The original train.csv contains five long-format rows per image; these are pivoted into a single five-output regression sample per image. Total samples after pivoting: 357.

### 2.2 State-Stratified Data Splitting

Pasture records exhibit significant geographical class imbalance across Australian states. To ensure robust evaluation, the data is split into Train (70%, n=249), Validation (15%, n=54), and Test (15%, n=54) sets using stratification on the State column. A fallback mechanism handles states with fewer than 3 samples to prevent splitting errors.

### 2.3 Feature & Target Normalization

Tabular features are z-score normalized using training-set statistics. Biomass targets are divided by a single shared global scaling factor (166.1 g, the max training target) to preserve the additive linear structure in normalized space. This is critical: if Total = Clover + Dead + Green holds in grams, it also holds after division by a common constant.

### 2.4 Model Architecture

The model is a two-stream multimodal fusion network: (1) a Visual Stream using MobileNetV2 (pre-trained on ImageNet) processes 160x160 pasture images, and (2) a Tabular Stream of dense layers encodes 24-dimensional metadata. The two embeddings are concatenated into a shared representation and fed through a 5-output regression head with softplus activation.

Softplus ($f(x) = \log(1 + \exp(x))$) replaces sigmoid and ReLU to avoid output saturation and dead neuron problems. The output bias is initialized with the inverse softplus of training target means ($b = \log(\exp(\mu) - 1)$) for stable early training.

### 2.5 Loss Function

Huber Loss (delta = 0.15) is used instead of MSE to handle the skewed target distributions and extreme outliers present in biomass data. Huber Loss behaves as L2 near zero and L1 for large errors, combining the stability of squared loss with the robustness of absolute loss.

### 2.6 Training Configuration

| Parameter | Value |
|---|---|
| Backbone | MobileNetV2 (ImageNet) |
| Image Size | 160 x 160 |
| Batch Size | 8 |
| Epochs | 50 (early stopping, patience=10) |
| Learning Rate | 0.001 |
| Dropout | 0.2 |
| LTN Weight | 0.01 |
| Rule Tolerance | 0.12 |
| LTN Warmup Epochs | 5 |
| LTN Ramp Epochs | 5 |
| Base Loss | Huber (delta=0.15) |
| Output Activation | Softplus |
| Seed | 42 |

## 3. Comparative Analysis

Three learning modes are compared on identical train/val/test splits:

### 3.1 Symbolic Baseline

Predicts primitive biomass components (Clover, Dead, Green) using group-mean estimates from metadata categories, then derives Total and GDM exactly from the additive equations. This baseline has zero rule violations by construction but limited predictive power because it ignores image evidence entirely.

### 3.2 Neural Model (Huber Loss Only)

Trains the full multimodal fusion network using only Huber regression loss. No logical constraints are applied. The model freely optimizes each of the five targets independently, which can produce physically inconsistent predictions (e.g., Total < Green).

### 3.3 LTN Model (Neuro-Symbolic)

Trains the same architecture with a combined loss: Huber regression plus a weighted satisfiability penalty from the LTN axiom knowledge base. The LTN loss is warmed up after 5 supervised-only epochs and linearly ramped over the next 5 epochs to prevent shortcut learning (collapsing all predictions to zero to trivially satisfy axioms).

## 4. Interpretation of Fuzzy Logics

The LTN framework translates classical first-order logic axioms into differentiable fuzzy operations over real-valued tensors. Each axiom produces a truth value in [0, 1], where 1 means fully satisfied and 0 means maximally violated.

### 4.1 Regression Predicates

A continuous regression predicate converts numerical agreement into a truth degree. For data fitting, truth is high when the predicted and observed values are close (relative to target-specific tolerances estimated from the training split). For rule matching, truth decays smoothly as the equation residual grows beyond the rule tolerance (0.12 in normalized space).

### 4.2 The Knowledge Base Axioms

The following axioms form the LTN knowledge base:

- Data Fitting: For all x, predicted values Close(f(x), y) should match observed targets.
- Total Mass Decomposition: Dry Total (g) = Dry Clover (g) + Dry Dead (g) + Dry Green (g).
- GDM Composition: GDM (g) = Dry Clover (g) + Dry Green (g).
- Non-negativity: All predicted masses must be >= 0.
- Part-Whole Ordering: Aggregate masses (Total, GDM) must be >= each component mass.

### 4.3 Satisfiability Aggregation

Individual axiom truth values are aggregated using the p-mean error operator (from the LTN reference implementation). The batch satisfiability score Sat(Phi) represents the overall logical consistency of predictions. The combined loss is: L_total = L_Huber + lambda_ltn * (1 - Sat(Phi)), where lambda_ltn = 0.01.

The soft nature of fuzzy logic is crucial for noisy biological measurements. A prediction with a small residual is not rejected; it receives a truth value slightly below 1. Only large violations produce low truth values and significant gradient pressure. This provides a smooth, differentiable signal that guides the model toward consistency without brittle hard constraints.

## 5. Results

### 5.1 Regression Performance (Test Set)

| Metric | Symbolic | Neural | LTN |
|---|---|---|---|
| MAE (g) - Overall | 8.75 | 10.69 | 9.31 |
| RMSE (g) - Overall | 15.15 | 15.59 | 15.37 |
| R2 - Overall | 0.470 | 0.440 | 0.347 |
| R2 - Dry Green | 0.636 | 0.455 | 0.632 |
| R2 - GDM | 0.506 | 0.570 | 0.668 |
| R2 - Dry Total | 0.511 | 0.551 | 0.451 |
| R2 - Dry Clover | 0.336 | 0.282 | 0.024 |
| R2 - Dry Dead | 0.363 | 0.340 | -0.040 |

### 5.2 Logical Consistency (Rule Violations, Test Set)

| Violation Metric | Symbolic | Neural | LTN | Reduction |
|---|---|---|---|---|
| Total Sum MAE (g) | 0.00 | 15.12 | 3.98 | 74% |
| GDM Sum MAE (g) | 0.00 | 11.26 | 4.84 | 57% |
| Negative Violation (g) | 0.00 | 0.00 | 0.00 | - |
| Order Violation (g) | 0.00 | 2.64 | 0.19 | 93% |

The LTN model achieves the best GDM R-squared (0.668) and matches the symbolic baseline on Dry Green R-squared (0.632 vs 0.636), while dramatically reducing rule violations compared to the pure neural model. Total sum violations drop by 74%, GDM violations by 57%, and ordering violations by 93%.

### 5.3 Training Dynamics


*Fig 1: Training and validation loss curves for Neural and LTN models.*


*Fig 2: LTN satisfiability score over training epochs.*

## 5.4 Predicted vs. Actual (Test Set)

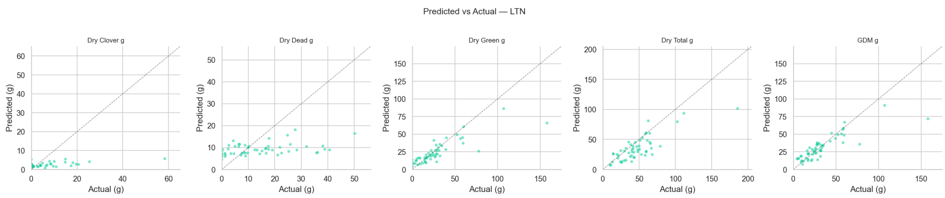


*Fig 3: LTN model - predicted vs. actual biomass (test set).*

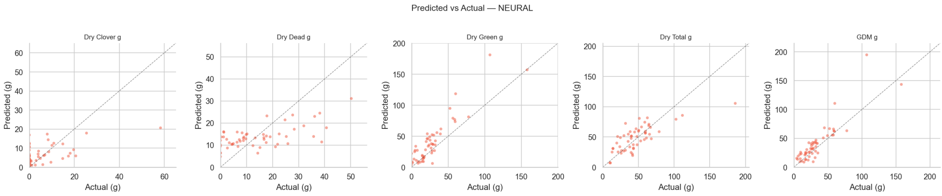

Fig 4: Neural model - predicted vs. actual biomass (test set).

### 5.5 Rule Violations Comparison


Fig 5: Rule violation comparison across all three modes.

Linear probe analysis: Neural probe R2 = 0.640, LTN probe R2 = 0.615. The LTN representation trades slight linear predictability for dramatically better constraint satisfaction.

## 6. Limitations

- Small dataset: With only 357 images, test metrics can vary noticeably depending on the random seed and split. Cross-validation would provide more stable estimates but was not implemented.
- Clover and Dead targets: The LTN model achieves near-zero or negative R-squared for Dry Clover and Dry Dead. Clover is approximately 38% zero-valued in the dataset, making it inherently difficult to predict. The logical constraints may be diverting capacity away from these sparse targets toward better consistency on the aggregate targets.
- Known-identity axioms: The logical rules encode known target identities, so the LTN layer improves consistency rather than discovering new biological laws. If labels contain rounding errors, strict tolerances can hurt supervised accuracy.
- Transfer learning sensitivity: MobileNetV2 ImageNet features were pre-trained on natural images, not agricultural plots. Domain-specific pre-training could improve visual feature quality.
- Shortcut learning risk: Without careful tuning of lambda_ltn and the warmup schedule, the model can collapse to predicting zero for all components (trivially satisfying sum constraints).

## 7. Directions for Future Work

- Cross-validation: Implement k-fold cross-validation stratified by State to reduce variance in performance estimates.
- Target transformations: Apply log or Box-Cox transforms to zero-inflated targets (Clover) to improve predictions for sparse components. Rules must then be evaluated in linear space by inverting the transform before computing axiom truth values.
- Learnable rule weights: Instead of a fixed lambda_ltn, learn per-axiom weights to let the model express confidence in each rule. This could prevent well-satisfied axioms from dominating the loss signal.
- Uncertainty estimation: Add Monte Carlo dropout or ensemble methods to quantify prediction uncertainty, especially for rare pasture types.
- Domain-specific pre-training: Fine-tune the visual backbone on agricultural imagery datasets before training on this

dataset.

- Calibration curves: Analyze how fuzzy predicate truth values correlate with actual prediction error to validate the semantic meaning of the satisfiability score.
- Ablation studies: Systematically disable individual axioms to measure the marginal contribution of each rule to overall performance and consistency.